

Fire Detection in Digital Images Using Enhanced Image Processing and Machine Learning Techniques

Daniel Winston Mandela Tulung (23/516260/PA/22080)*,
Girindra Daafi Mada (23/511637/PA/21835)[†],
Keefani Mentari Ingprairie (23/511420/PA/21792)[‡],
Naura Ayesha Tsaqif (23/511457/PA/21800)[§]

^{*†‡§}Department of Computer Science and Electronics, Universitas Gadjah Mada,
D. I. Yogyakarta, Indonesia

^{*}danielwinstonmandela@mail.ugm.ac.id, [†]girindradaafimada@mail.ugm.ac.id,
[‡]keefanimentariingprairie@mail.ugm.ac.id, [§]nauraayeshatsaqif@mail.ugm.ac.id

Abstract—The quick detection of fire is crucial in minimizing economic and human lives lost. This paper examines fire detection using a forest fire dataset based on a paper by Wahyono et al. [1] and intends to create effective and scalable methods for reducing fire-related hazards with the proposed model. The approach uses advanced image processing techniques such as Fire Mask Thresholding, Otsu Mask Thresholding, and Sobel Edge Detection which are then utilized as input to a Support Vector Machine (SVM) algorithm.

Keywords: Fire detection, Support Vector Machine, fire monitoring, hazard mitigation, fire mask thresholding, otsu mask thresholding, sobel edge detection.

I. INTRODUCTION

Fires can start quickly, especially in environments with enclosed or semi-enclosed environments that pose high risks of fire due to factors such as faulty electrical wiring, unattended cooking or simply carelessness in handling flammable materials. In residential areas or places where people gather, an example is when poorly maintained electrical systems cause fire in apartments and office buildings. With a single spark, an undetected flame can lead to a larger fire. Similarly, cooking-related incidents are also common; Without quick action, the fire will eventually spread. Not to mention, the amount of gas in the area could make it even worse.

On a larger scale, fire hazards in open environments such as fields, plantations, and forests must be strictly supervised. Indonesia's dry seasons have historically led to widespread forest fires, worsened by illegal land clearing methods and the extensive use of peat lands, which are highly flammable [2]. Once started, these types of fire spread quickly due to dry conditions and wind, making them difficult to control. This results in long-term ecological damage, economic loss, as well as poor air quality. Furthermore, smoke from forest fires has the potential to cover large swathes of land and release millions of tons of carbon dioxide into the environment [2].

II. RELATED WORKS

Currently, our work is based on several studies done on fire recognition to acquire various methods on its detection. Firstly, Rahman et al.'s study on forest fire detection using Support

Vector Machine(SVM) [3] shows a 93% accuracy rate when comparing their model with a model utilizing SVM with a lab histogram and SVM with motion detection and color detection. However, this study detects fire from videos and we aim to detect fire from images alone. Secondly, study from Chen et al.

[4] provides the basic idea of segmentation using RGB values. Their model detects the red value of the image and if said value surpasses a certain threshold and that if the red values are also greater than the green and blue values, it would detect fire. Furthermore, supplementary processes, like the growth of fire, are utilized when the first process isn't enough to detect the fire. However, using RGB values instead of HSV values tend to produce skewed results as shadows in RGB images have different characteristics than the parts that are not shadows. Finally, Wahyono et al. [1] uses machine learning techniques like Support Vector Machines and Random Forrest to verify whether the image includes fire or not. Unlike Rahman et al.'s work, this paper mainly focuses on forest fire detection and on an early warning system as it detects the fire. Furthermore, this paper has become the backbone of our work as we aim to improve on the techniques while also implementing various other techniques that can improve the results which will be discussed further in this paper.

III. METHODOLOGY

Our proposed solution utilizes the AESFERM framework of digital image processing to detect fire in images. The main goal is to correctly and effectively identify fire in images with various background and minimize false positives caused by similar phenomena such as sunlight or smoke. This structured approach improve proven techniques from digital image processing and feature engineering to achieve precise and reliable results.

To implement the proposed solution, a few libraries are implemented such as: 'random, os' to manage file paths and select random images, 'cv2(OpenCV)' to read, process and manipulate images, 'matplotlib.pyplot' to display images, 'numpy' for numerical operations like image manipulation and feature extraction, 'pandas' to store extracted features in a DataFrame and save them as a CSV, 'skimage.feature'

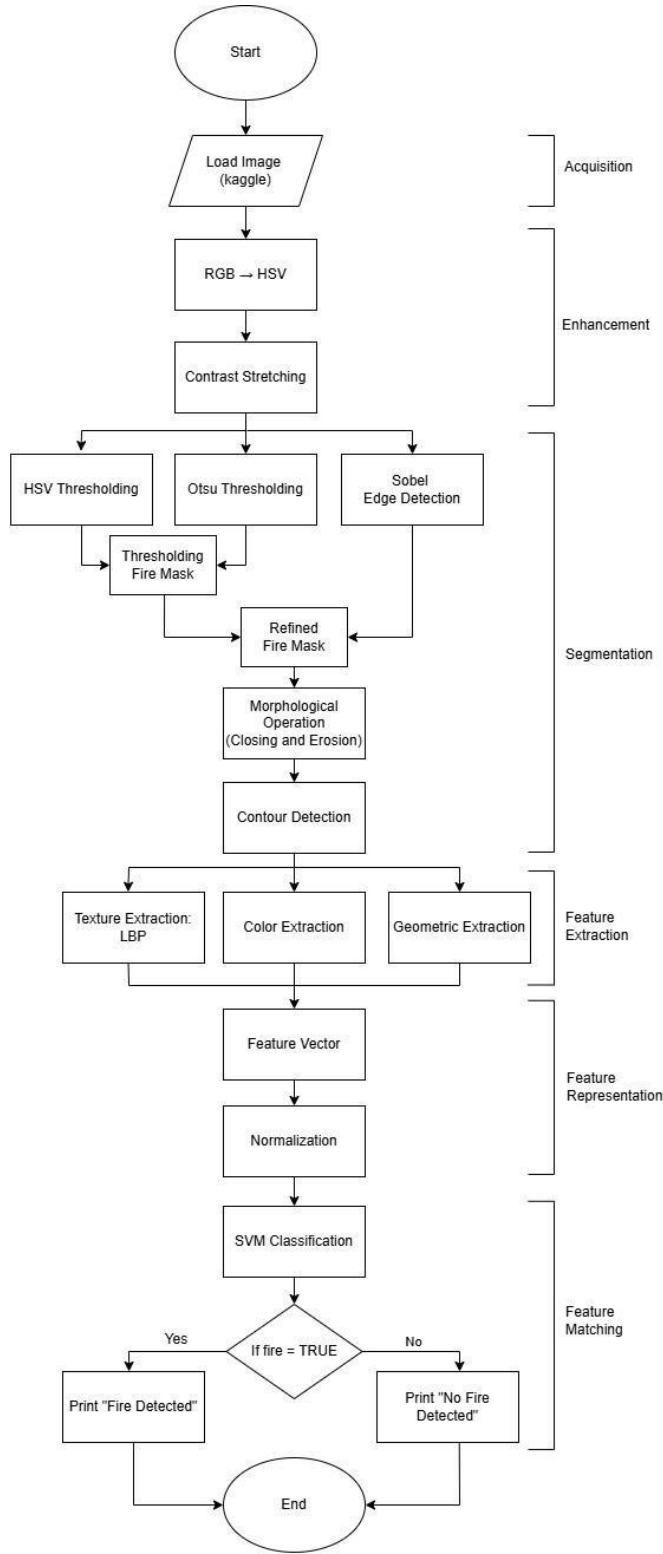


Fig. 1. Flowchart of the model

which provides the 'local_binary_pattern' function for texture analysis, 'scikit-learn' for model training, splitting data, and

standardizing features, and lastly 'joblib' for saving the trained model and scaler. Subsequently, the following AESFERM techniques are applied:

A. Acquisition

In the early phase of the system, we focus on loading preprocessed images into the system to make sure that the data we work with is optimal for processing. Here, we gathered 760 fire images and 760 non-fire images from a publicly available Kaggle dataset. The dataset used in this project contains images of forests, with some images featuring the fire and others with fire-like regions or without. This dataset is reliable as the images are representative of real-world scenarios where fire might occur, capturing varying lighting and weather conditions, making them ideal for fire detection tasks. The images are downloaded from the Kaggle dataset and stored locally for easy access during the preprocessing and processing stages.

B. Enhancement

Enhancement phase aims to prepare the images for effective extraction later on in the process and also the detection of fire itself. By upgrading the visibility and contrast of fire-related color, we are going to create a dataset that allows for better analysis. Moreover, enhancing the image ensures that the subsequent segmentation techniques have high-quality data to work with, ultimately improving detection accuracy. Our proposed solution uses the two following enhancement techniques:

- 1) *Color space conversion:* This is where the RGB images are converted to HSV. This is important because it separates the image into three components: Hue, Saturation, and Value. As illustrated in figure 2, Hue represents the color type (where 0° represents red, 120° represents green, and 240° represents blue), Saturation represents the intensity of color or how vivid the colors are (where 0 means the pixel is gray and 255 means the color is the most vivid; no gray is mixed with the color), and Value represent the brightness (where 0 means the pixel is black and 255 means the pixel is at its maximum brightness).
- 2) *Contrast stretching:* Next, contrast stretching is applied to the Value (V) channel. Because the Value channel represents the brightness, contrast stretching is done to this channel to improve the visibility of fire-related colors to make it easier for the segmentation step to differentiate between fire and non-fire elements. Contrast stretching causes the bright colors (fire-related colors) to be brighter and dark colors (non fire related colors) to be darker. This step is crucial as raw images often have varying quality due to environmental factors like lighting and weather.

C. Segmentation

The main objective for the proposed segmentation techniques is to isolate regions of interest i.e. the fire regions,

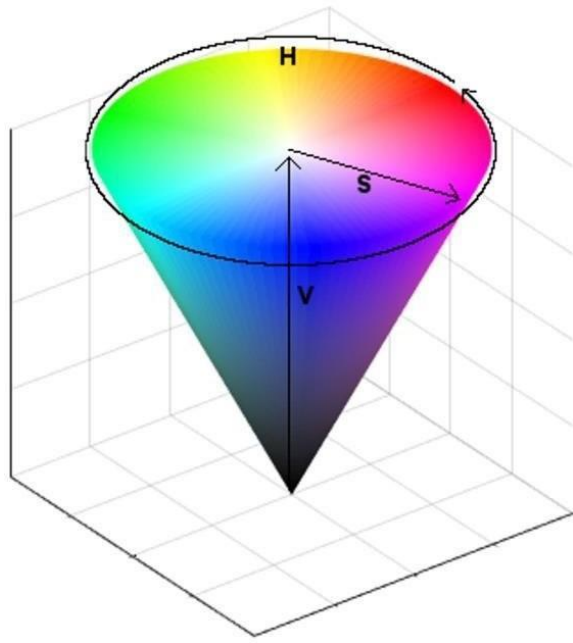


Fig. 2. Visual Representation of HSV

from the background in the image effectively. Color segmentation, Otsu's thresholding, edge detection, and morphological operations were combined to achieve our main goal.

- 1) *Color Segmentation*: After the above enhancement techniques are applied, a binary fire mask is created using the following HSV threshold: 10, 100, 100 for the lower bound and 20, 255, 255 for the upper bound. The stated Hue threshold was chosen to isolate the red-orange-yellow colors just like the typical fire colors. The Saturation threshold was chosen to be 100-255 as fire usually have a vivid color and has high intensity. Lastly, the chosen 100-255 Value colors is to ensure that the bright pixels are isolated from the dark or non-fire pixels. At this stage, the fire mask will consist of white (255) pixels which represents pixels that fall within the range and black (0) pixels which represents pixels that are not within the range.
- 2) *Otsu's Thresholding*: Otsu's Thresholding was implemented next to eliminate possible false positives from objects like sunset or red/orange objects that were identified from color segmentation alone. This method is applied globally to the Value channel to isolate areas with bright intensities more effectively. This method was chosen as it automatically calculates the optimal threshold value, which is effective for separating fire pixels from the background. It distinguishes the pixel intensities into two classes: fire and non-fire. The pixels that are above the threshold, the identified potential fire region, will be white (255) and the pixels that are below the threshold, the background, will be black (0). Furthermore, the advantage of using Otsu's algorithm is

that it is suitable for different lighting conditions as it dynamically adapts to different images.

The binary mask created based on the Otsu's thresholding is combined with the previously made fire mask using the AND bit-wise operator. At this stage, the mask is refined as only regions that have fire-like characteristics in terms of colour and brightness are retained.

- 3) *Edge detection*: Sobel Edge detection is used to identify boundary between fire (region of interest) and the background. It highlight regions with strong intensity changes which are usually object boundaries while also smoothen the image. Moreover, the jagged and irregular contours characteristic of fire can be identified using this technique. It is applied to the Value channel of the enhanced HSV image.
Two 3x3 convolution kernels are used: one to calculate the horizontal gradient component and the other to calculate the vertical gradient component. Then, the two gradients are calculated to obtain the overall gradient magnitude. Just like in the previous steps, the bitwise AND operator is used again to refine the fire mask. Now, only the region with sharp edges remain in the fire mask.
- 4) *Morphological operations*: This segmentation method consist of two operations such as closing (combination of dilation and erosion) and erosion. A 5x5 kernel was used to implement the two operations. The closing operation was done to connect separated and fragmented fire regions by adding white pixels to the edges and then remove extra noises created by dilation. An additional erosion operation is done to remove more of small noises in the image to ensure the image is smooth. By the end of the three segmentation method, we will obtain an isolated fire-like regions with no errors caused by noise or environmental interference.
- 5) *Contour Detection*: The last segmentation method helps identify the boundaries of individual fire-like regions that need to be processed and analyzed further. This method allows us to only focus on the outermost contours, ensuring that we capture the boundaries of the primary fire areas and ignore any internal details or nested contours. This is especially important in fire detection because we're interested in analyzing the overall shape, size, and position of the fire - not internal fluctuations. Moreover, the use of contour detection can assist in calculating critical geometric and shape-based features such as area (the size of the fire region), perimeter (the length of the boundary, providing information about the fire's shape complexity), and aspect ratio and circularity (metrics that determine whether the region resembles typical fire patterns). By analyzing these features, we can filter out irrelevant or noise-based regions (e.g., small reflections or isolated bright spots) and focus only on regions that match fire-like characteristics. Overlapping regions in the fire mask can also be separated and so we do not treat multiple fires as a single object. Lastly, localizing fire regions can be done precisely to track

fire spread, estimate its impact or direct firefighting resources to specific areas.

D. Feature Extraction

There are several features to be extracted from the image after enhancement and segmentation. The purpose of this step is to extract relevant features from the segmented regions to characterize fire regions and non-fire regions. The features extracted are:

- 1) *Geometric Features*: This include extracting area, perimeter, aspect ratio, extent, and circularity from detected contours. These features are helpful in distinguishing fire regions from non-fire regions/objects with similar colors but distinct shapes. For instance, traffic lights or orange signage.
- 2) *Color Features*: This is done by calculating the mean values of the HSV channels within the fire region to capture the overall color characteristics of fire.
- 3) *Texture Features*: This is done by utilizing the Local Binary Patterns technique. LBP can reveal information about the fine grained-details of fire texture patterns. Typically, fire possess a complex, turbulent textures due to the motion of flames and heat and LBP is the perfect method to identify them. These complex textures are useful to distinguish fire from non-fire regions that have similar colors but lack the characteristic textures of flames. LBP works by comparing pixel intensities in a small neighborhood around each pixel and encoding this information as binary patterns. This helps the system distinguish flames from smooth surfaces or flat areas that do not exhibit such texture. Furthermore, LBP focuses on the relative intensity differences between neighboring pixels, rather than the absolute pixel values. This makes it particularly useful for fire detection in varying lighting conditions, where the flame textures might still be distinguishable despite the light variations. Not only it is felicitous in removing patterns, it also helps with SVM classification that will be applied later on in the process.
- 4) *Gradient-based Features*: This is done by calculating the mean and standard deviation of gradient magnitudes in the x and y directions that was done using Sobel filters in the Segmentation stage. This helps in capturing texture and intensity changes within the fire region.

E. Feature Representation

In the second last stage of the process, the extracted features are converted into a suitable format for machine learning or comparison. The two technique we applied are:

- 1) *Feature vector*: This method combine all extracted features such as shape, color, edge, gradient, and texture into a single feature vector for each detected fire region.
- 2) *Normalization*: The feature vectors are then normalized to ensure that all features contribute equally to the model's learning process

F. Feature Matching

The last stage of this process focuses on detecting the fire from images by comparing the known characteristics with the characteristics found on the image inserted. The Feature Matching technique proposed is using Support Vector Machine (SVM). This technique was chosen as it learns patterns in the extracted features and uses the information to make predictions on new image input. Using the feature vectors (color, texture, shape) from the extraction stage, it became the input to classify as either “fire” or “non-fire”. During training, SVM algorithm learns the relationship between feature vectors and the label (fire/non-fire) and during testing, SVM uses the feature vectors from the new image to predict whether they contain fire [5].

G. Visualization and Post-processing

Finally, to clearly show that fire is detected on an image, bounding boxes are drawn around the fire. Otherwise, nothing is drawn on the image and the output will state that fire is not detected on the image.

IV. EXPERIMENTAL RESULT AND DISCUSSION

The Google Colab file can be accessed through this link: <https://colab.research.google.com/drive/1SdSj29U3FlXWDCILWYtubfoGDaMHR3?usp=sharing>

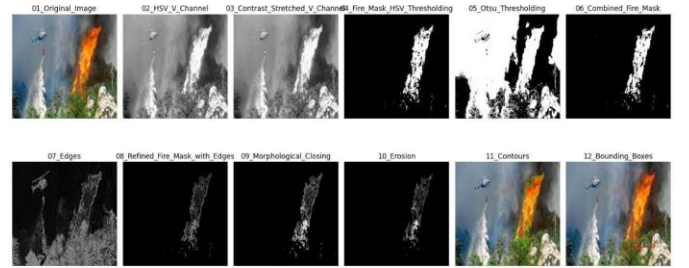


Fig. 3. Process of fire detection

In figure 3, we firstly import the image from our database and convert it into HSV for the preprocessing. Secondly, the HSV image is then contrast stretched and then the model does two types of segmentation for feature extraction, which are fire mask and Otsu segmentation. Afterwards, the two are merged into 1 to detect the fire by taking what both the fire mask and Otsu segmentation detects as fire. Next, we use edge detection from the contrast stretched HSV image and merge it with the previously joined segmentation to refine it once more. Moreover, we use morphological closing and erosion to fill in the gaps from detecting the fire. Finally, the contours are drawn and the boxes are drawn from the contour to detect the fire.

The dataset used is split by 80% for training and 20% for testing. Furthermore, 5 randomly selected images from both the fire and non-fire image dataset are chosen during a single round of testing. Figures 4-6 are randomly-picked results and from it, it was able to correctly identify most fire images with a true positive. However, there are also some pictures, like in

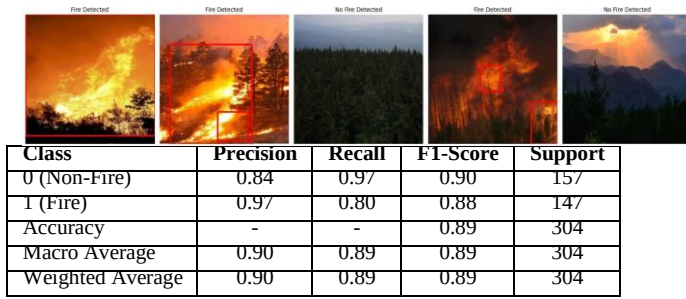


Fig. 4. Result of fire detection 1



Fig. 5. Result of fire detection 2



Fig. 6. Result of fire detection 3

Figure 6, that is a fire image but was not detected. The model tends to struggle when there is smoke or haze in the image.

After the model has been trained and tested, the model is predicted have an accuracy of 89.14%, which indicates that the combination of techniques we have utilized for our model is fairly accurate. Furthermore, this model has successfully shown that it can reliably not detect fires on fire-like images as shown in Figure 8. However, several problems arise from this model that are shown in Figure 7, which are that the recall for the fire images are only around 80% and that the precision of the non-fire images are also only about 84%. Firstly, the model will falsely flag a fire image as a negative 20% of the time. This should be one of the first things to be improved

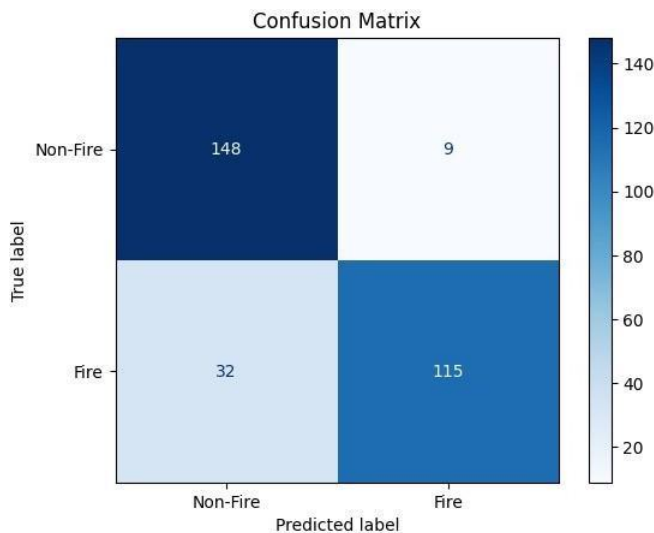


Fig. 7. Confusion Matrix

TABLE I
RESULT OF CLASSIFICATION RESULTS WITH ACCURACY OF 89.14%



Fig. 8. Positive result from fire-like images

upon when a new iteration is made as it is very dangerous if a fire goes undetected and can lead to loss of life or economic loss. Secondly, the model has an 16% chance to flag non- fire images as positive. While this problem isn't as serious as the previous one, it could also lead to wasted resources and economic

V. CONCLUSION

The danger posed by uncontrollable fire should not be discarded lightly. Small sparks can lead to huge burnings which could in turn cause economical damage, ecological damage, and even death. In this study, we based our model and research off a study from Wahyono et al. [1] and from our results, we have achieved an accuracy similar to the model proposed by Wahyono et al. with an accuracy of 89.14%. However, the model still struggles to correctly identify fire images that features a lot of smoke, leading to false negatives which should be one of the first problems to look into when revisiting this study.

REFERENCES

- [1] Wahyono, A. Harjoko, A. Dharmawan, F. D. Adhinata, G. Kosala, and K.-H. Jo, "Real-time forest fire detection framework based on artificial intelligence using color probability model and motion feature analysis," *Fire*, vol. 5, p. 23, 2 2022. [Online]. Available: <https://www.mdpi.com/2571-6255/5/1/23>
- [2] Xinhua, "Indonesia braces up for alarming forest fires during dry season - global times," 8 2024. [Online]. Available: <https://www.globaltimes.cn/page/202408/1317386.shtml>
- [3] M. A. Rahman, S. T. Hasan, and M. A. Kader, "Computer vision based industrial and forest fire detection using support vector machine (svm)," in *2022 International Conference on Innovations in Science, Engineering and Technology, ICISSET 2022*. Institute of Electrical and Electronics Engineers Inc., 2022, pp. 233–238.
- [4] T.-H. Chen, P.-H. Wu, and Y.-C. Chiou, "An early fire-detection method based on image processing," 2004.
- [5] M. A. I. Mahmoud and H. Ren, "Forest fire detection and identification using image processing and svm," *Journal of Information Processing Systems*, vol. 15, pp. 159–168, 2019.